



Description and operating instructions

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1. Introduction

**The small powerful RC data logger and telemetry sensor
Stores and transmits
V / A / W / mAh / Wmin / rpm / ° C / Single cells / Height / Vario**

The UniLog 2 is the successor of our successful data logger the UniLog. All well known functions of the UniLog were carried forward with current technology of the latest type. Numerous new functions have been added so that practically nothing is left to wish for.

Telemetry over 2,4 GHz systems with a back channel is fully integrated and is now component of the UniLog 2.

In line with our philosophy to support as many systems as possible, the UniLog 2 also speaks via the telemetry:

Jeti Duplex EX

Multiplex M-Link

Graupner/SJ HoTT

Robbe/Futaba S.BUS2

JR DMSS

FrSky

Spektrum

PowerBox P2Bus

The telemetry used must be specified only once in the in UniLog 2 settings. This happens Either via our PC program "UniLog 2 Tool" or with the UniDisplay. In delivery condition HoTT GAM is selected.

Since an additional barometric (air pressure) altitude sensor with high resolution is also integrated, a Vario is realized in the UniLog 2 with the telemetry in use. Likewise extensive alarms are programmable, which are announced by telemetry over the respective transmitter.

When operating with Multiplex M-Link all data on the sensor bus is automatically logged by the UniLog 2 and also written to the memory card ➔ MSB Data logger.

A micro SD memory card provides the UniLog 2 with practically unlimited memory. With up to 20 cycles per second recording rate, plus the micro SD memory card, arbitrarily long recordings with high resolution and detail are possible. Data exchange with the PC for evaluation is also very easy thereby: take the memory card from the UniLog 2 and with the card reader provided put it into a PC USB port. The measured values of the UniLog 2 are present as text files on the memory card.

The data is then displayed and evaluated using the GNU DataExplorer (www.nongnu.org/dataexplorer/index.en.html) or with LogView (www.logview.info). With the internal real-time clock with battery, all readings are always stamped with date and time.

Also new is the possibility of single cell voltage measurement. A LiPo up to 6s can be attached directly via its balancer plug to the UniLog 2, and so the voltages are measured and recorded and naturally also transmitted by telemetry.

Via our UniDisplay all values measured by the UniLog 2 can be viewed live and also the data read from the memory card. Maximum and Minimum values can quickly be called up from the stored recordings. Naturally all settings and alarms can also be programmed easily by UniDisplay.

Whether sailplane, aerobatics plane, helicopter, HLG or Slowflyer, the UniLog 2 can be used in almost any sector due to its small weight and compact size. Naturally the UniLog 2 is not only suitable for the Model flying, it can also be used in RC-boats, RC-car etc.

2. What the UniLog 2 can do

- Complete measurement of electric drives with current, voltage, power, capacity, number of revolutions (rpm) ,and temperature measurement (current, rpm and temperature sensors are separately available in different variants)
- Micro SD memory card → almost unlimited recording and simple selection of data
- Full telemetry support for Jeti Duplex, Multiplex M-Link and Graupner HoTT, Futaba S.Bus2, JR DMSS, FrSky, Spektrum, and PowerBox P2Bus.
- Internal real-time clock (with battery for at least 10 years) → all measurements will be provide with real time stamp of date and time
- Memory rates of 1 measurement per second (1 Hz) to 20 measurements per second (20 Hz)
- All sensors of the UniLog can still be used
- Balancer connector for LiPo single cell measurement up to 6s
- Altitude with automatic zeroing after switching on
- Current measuring range depends upon sensor used up to 400 A, voltage up to 60 V
- Current and voltage sensor plugged in (so easily used as a small pure altimeters/Vario e.g. in a Sailplane)
- Connection for receiver signal (Rx) and signal output to the automatic controller (ESC)
- Universal connections for external rpm sensor and up to three temperature sensors
- Temperature measurement also with PT1000 sensors for an extended measuring range
- Connection for speed sensor (stock no. 2560) for measuring airspeeds
- Direct connection to the GPS Logger is possible for data capture in a file
- Recording of all data on the Multiplex Sensor Bus in operation with M-Link
- Recording of the receiver voltage
- Power supply automatically from receiver and/or motor battery
- Manual recording start by integrated button
- Start of the recording by different set conditions
- Current status is indicated by three LEDs
- Direct viewing of recorded or live values with our UniDisplay
- Parameter settings over PC, UniDisplay, or telemetry is possible
- Support by LogView software www.logview.info
LogView is a very comprehensive yet easy to use evaluation programme for the PC that supports a multiplicity of different measuring devices and battery chargers from the model construction field.
- Supported by the GNU DataExplorer software www.nongnu.org/dataexplorer
The GNUDataexplorer is just as extensive as LogView and, in addition to Windows, is also suitable for Mac and Linux.
- Simple and clear evaluation of measurement with the aid of diagrams on the PC
- Free firmware updates possible via the memory card (the firmware file is available on the Internet at www.sm-modellbau.de in the menu option **Software & Updates**)
- Useable almost everywhere due to its compact size and low weight

3. Technical data

Data Rate	1 Hz, 2 Hz, 5 Hz, 10 Hz, 20 Hz adjustable.
Memory Type	Micro SD or micro SDHC card (supplied with 2 GB card)
Recording duration	with 20 Hz data rate and full extent of utilization approx. 240 kByte/minute storage requirement → nearly 6 days recording with 2 GB card with 20 Hz data rate → nearly 12 days with 2 GB card with 10 Hz data rate → nearly 4 months with 2 GB card with 1 Hz data rate
Current measuring range	Depending upon sensor 20 A, 40 A, 80 A, 150 A or 400 A, also negative currents.
Voltage measuring range:	0 to 60 V
Single cell measurement	Up to 6s LiPo, each connection max. 28 V
Altitude measuring range	0 to 8000 m NN, when switching on receiver altitude measurement is zeroed automatically
Receiver battery voltage range	0 V to 10 V
Current supply	from receiver supply over the telemetry connection "Link" (from 3,8 V to maximally 10 V) or directly over the current and voltage sensor from the motor battery (with more than 8s or 35 V the UniLog 2 must also be supplied from the receiver)
Current consumption	approx. 44 mA
External connections	1 x combined current and voltage sensor 1x pin row for balancer connection to 6s LiPo 1x rpm sensor 3x universal analogue lines for temperature sensor, speed sensor etc. 1 x servo impulse input from the receiver (Rx) 1 x servo impulse output to the controller (ESC) GPN servo cable for current supply and/or telemetry COM connection for UniDisplay, GPS Logger, or PC Card location for micro SD card
Dimensions	42 x 25 x 10 mm
Mass	9 g without cables

4. Data Capture

The UniLog 2 can capture the following values. Most of the data is stored on the memory card and at the same time transmitted by telemetry. Only part of it may be available at the transmitter depending on the telemetry used.

Description	Unit	Content
Drive current	A	Current in amperes taken by the drive at up to 2 decimal places depending on telemetry system.
Drive voltage	V	Voltage of the drive battery in volts at up to 2 decimal places depending on telemetry system.
Capacity	mAh	Battery capacity consumed by the drive, with a fully charged battery is incremented from 0 mAh.
Energy	Wmin	Energy consumed by the drive in Watt minutes.
Power	W	Power taken by the drive in Watts.
Revolution speed	rpm	On a speed sensor connected to the rpm input
Cell voltages	V	Voltage of up to 6 individual cells via the balancer connector
Altitude	m	Height above the start point measured from the barometric sensor.
Climb rate	m/s	measured from the barometric sensor.
Height gain	m	Height change in the last 10 seconds, recalculated every second so the trend can be recognised when thermaling.
Receiver volts	V or VRx	Voltage at the power supply input of the UniLog 2
Servo impulse in	us	Measured servo pulse at the input Rx. Can optionally be used for switching between Min/Live/Max values or in Limiter operation.
Servo impulse out	us	Servo pulse issued from port "ESC", is used in Limiter operation.
Pressure	hPa	Measurement of barometric air pressure sensor
Temperature	°C	Temperature sensor connected to one of the inputs A1 to A3
Speed	km/h	Speed sensor connected to one of the inputs A1 to A3
Remaining capacity	%	• Residual charge of the drive battery calculated from the set capacity alarm value and the mAh consumed by the drive. • Example: 2200 mAh battery, alarm value set 1760 mAh for 80% discharge ➔ Residual capacity is 100% when the battery is full and 0% when the 1760 mAh has been consumed.

5. Operation of the UniLog 2

5.1. Installation

Due to the low weight and the compact design, installation is unproblematic. Attention must be paid only to the fact that the start/stop key cannot be inadvertently pressed. Attachment with a Velcro strap on a board is sufficient and makes easy access possible to the memory card.

5.2. Memory cards

As memory card practically all commercial micro SD cards with FAT16 or FAT32 file system can be used. Also SDHC cards and memory capacities over 2 GB are supported. However not all cards are equally suitable, since some cards exhibit an unfavorable behavior with continuous storing of data. If an unsuitable card is used, the recording may run intermittently or even stop. We recommend the use only with the card provided or available as an accessory from us.

The card is inserted into the cut-out on the front and pushed in until it clicks and is flush. The UniLog 2 does not have an ejector for the memory card, the card is simply pulled out again with the fingers.



5.3. Meaning of the LED

The UniLog 2 has three coloured LEDs.

After switching on a run of the three LED indicates the internal initialization.

In the operation there are the following signals:

- **green LED shines permanently**
→ UniLog 2 ready
- **green LED flashes at the set recording rate**
→ UniLog 2 is recording data
- **red LED flashes slowly**
→ no memory card pushed in.

5.4 Basic Settings

The settings of the uniLog 2 can be made with a PC or laptop equipped with our software "SM UniLog 2 Tool", with our UniDisplay, or with the HoTT and Jeti telemetry.

The settings are always backed up in parallel on then UniLog 2 and saved on the memory card. When new settings are created using the PC software "SM UniLog 2 Tool" and have been saved on the card, they will be loaded by the device at the next start.

→ In this way it is possible to use different memory maps for different models and to automatically receive the correct settings.

The following settings are important so that the UniLog 2 can function correctly:

- **"Telemetrie Auswahl"** specifies the telemetry used. With HOTT, the "HOTT Mode" is also set here.
- **"fixe Seriennummer"** so that Jeti and Futaba sensors can be interchanged
- **"Datenrate"** selects the recording speed
A reasonable value is 5 or 10 Hz here, so the files are not too big and yet all details are recorded.
- **"Sensor"** selects the connected current sensor (20 A / 80 A / 150 A / 400 A.
Here, the actual connected power sensor must be set so that the UniLog 2 can measure the current correctly.
- **"A Modus"** selects the sensor connected to these analog inputs.
The selection must correspond to the actual sensors connected, or false readings may be displayed.
- **" Propellerblätter "** sets the number of propeller blades in the optical and the number of magnets in the fixed magnet speed measurement.
- **"Pole"** sets the number of magnetic poles in the brushless speed measurement (typical inner rotor 2 poles External rotor 10 or 14 poles)
When in doubt, this value must be obtained from the manufacturer, or the magnets counted.
Propeller blades and poles are set together, there are no independent values.
- **"Gertrieb"** is the gear ratio for the speed measurement.
When in direct drive or for the speed measurement on propeller or on the drive shaft "1.00:1" must be set here. Otherwise, the value of the subsequent transmission is specified.
- **"Vario Schwelle"** is the threshold for the vario signal via telemetry.
Only when the climb / descent is greater than the threshold, is a Vario tone generated by telemetry.
- **"Vario Ton"** sets whether the Vario tone is active during climb / descent or both.
The Vario tone can also be switched off completely here.
- **"Vario Filter"** allows an adaptation of the response speed of the varios.
"Slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster.
At the same time, however, the changing signal of the varios is always more rapid
- **"Vario bei motor"** determines whether the Vario is active while the engine is running
- **"Limiter modus"** defines the competitive class for the limiter.
Thus is selected a special operating mode for competition classes which require a power limiter. In normal operation, "Limit Mode" is set to "off".
- **"Energie Limit"** is the corresponding energy limit in Wmin.
- **"Zeit"** und **"Datum"** are used to set the internal clock.
- **"Kapazität"** sets the mode for the capacity measurement: continuous or always start at 0 mAh. (see below).
- **"Stromoffset"** specifies whether the when you connect the drive battery to UniLog 2 the actually flowing current should always calibrate to 0.
Thus, the quiescent current of servos, etc. is hidden.
- **"Rx Eingang"** determines the use of the Rx input.
Min / Live / Max allows switching to the current Min and Max values with a free receiver channel.vThese values are then transmitted via telemetry instead of the live values.

Capacity measurement options:

Setting "fortgesetzt"

With the UniLog 2 a battery can be flown empty in several flights. Here, the UniLog 2 notes the capacity (and energy) used and starts again with this value, unless a battery is connected at full voltage (i.e. fully charged)

Einstellung „immer bei 0 starten“

The removed capacity is not saved and always starts at 0.

Einstellung „immer weiter“

The removed capacity is stored and incremented at each new start. Only a manual reset to 0 is possible here.

Manual reset of the counter capacity:

Start UniLog 2 and switch off during the fast flashing (flicker) of the red LED

→ the counter is set to 0 again next time.

Or for manual reset of the min/max values by the receiver channel, see section 7.5 "Min and Max values per switched receiver channel".

5.5. Telemetry Alarms

These alarms are output via the telemetry connected to the transmitter. Depending on the system, a beep will sound and / or a warning by voice output. Please read the notes in the individual telemetry system manuals.

Once the model has landed the acoustic output stops automatically, so turn off the model and no more disturbing messages are issued.

"Strom" (current)

The alarm is active when the set current threshold is exceeded.

"Startspannung" (Start Voltage)

This alarm serves as a warning before the starting with an empty battery. For example if 12.4 V is set for a 3s LiPo, the alarm is only active if by mistake an already flown empty battery is connected. A full 3s has about 12.6 V, which is significantly above the threshold.

"Spannung" (Voltage)

The alarm is active when the set voltage threshold is dropped below.

"Kapazität" (capacity)

The alarm is activated when the consumed capacity exceeds the set value.

This is the most important alarm to conserve the LiPo. A max value of 80% of rated capacity should be set.

"Energie"

The alarm is activated when the energy consumed exceeds the set value.

"Hohe" (Altitude)

The alarm is active when the set height level is exceeded. Well suited to towing a plane to a certain level.

"Rx Spannung" (Rx Voltage)

To monitor the receiver supply. The alarm is active when the set voltage threshold is dropped below.

"Zellenspannung" (Cell Voltage)

The alarm is active when at least one of a maximum of 6 cells connected to the balancer falls below the set voltage threshold.

"A1-A3"

Freely configurable alarms for measured values at the analog inputs A1 to A3. It can either be the alarm value and the direction set. Thus, for example an alarm can be triggered when a temperature is exceeded by a temperature sensor or a speed with a speed sensor drops below a set value.

"Drehzahl min und max" (rpm min and max)

To monitor the measured speed. The alarm is active as long as the value exceeds or falls below the set value.

5.6. Start and stop the recording

The UniLog 2 has several possibilities for starting data recording. The appropriate options can be set over our software "SM UniLog 2 Tools", over the UniDisplay or over the Jeti and HoTT telemetry.

With an electric drive in order to automatically record every flight in a new file, we recommended starting by current threshold, e.g. with 3 A drive current and automatic stop of recording after landing.

The recording of the data can be started / stopped in the following way:

- **Manual start/stop with the button:**
The recording is started and stopped by depressing the button on the UniLog 2.
- **Manual start via telemetry:**
The recording is started and stopped in the Jeti Duplex or Graupner HoTT telemetry (text mode) by depressing a key at the transmitter.
- **Start with exceeding an adjustable current threshold:**
The recording begins automatically, as soon as the measured current is larger than the threshold. In this way the recording can be started with the start of an electrical model.
- **Start at the end of an adjustable time:**
If "Autostart time" is activated, the recording begins automatically as soon as the set time is completed after the start. So you do not have to worry about pressing the button, the recording starts itself.
- **Start/stop with remote control signal (servo impulse):**
If the option "Rx start" is activated, the recording starts as soon as the impulse from the receiver exceeds the set threshold. If the impulse drops below this threshold then the recording will stop again.
In this way the recording of the UniLog 2 can be controlled by the transmitter with a two position switch via a free channel. The switch must be defined in such a way that the free channel shifts up and down between -100% (recording stop) and +100% (recording start).
The value 1.5 ms corresponds to the central position of most current remote controls. For special applications the asynchronous operation threshold can be adjusted from 1.1 ms to 1.9 ms.
The value "Rx an" starts the recording, as soon as the UniLog 2 receives a valid receiver signal. In this way e.g. in competition models where the motor battery is connected previously, the recording starts as soon as the receiver is switched on.
- **Automatic stop after landing:**
With the option "Autostop landing" the recording ends automatically 30 seconds after the landing, that is if for 30 seconds the height is less than 10 m and the drive current less than 5 A.

All starting options can be active at the same time. In this way e.g. if the recording was started by current, the recording can be stopped again with the remote control signal. The switch at the transmitter must be moved only once from +100% to -100%.

With each start of recording the UniLog 2 begins a new file. The file names are sequentially numbered and have the following format: **"2017-01-01 SM UniLog 2 Datei 0001.txt"**

To distinguish between different firmware versions these files are always in a folder with this format: **"SM UniLog 2 v1.13"**

In each case the recording should be stopped before disconnecting the UniLog 2, since only that way are the minimum and maximum values stored for use with the UniDisplay. The recording can also be terminated simply by interruption of the supply current. That is intended and OK, however the minimum and maximum values in the file are then missing.

5.7 Vario output via telemetry

The Vario tone generation is carried out in all cases (with the exception of the "old" Jeti non EX telemetry) in the telemetry transmitter or the telemetry box. For this purpose, the UniLog 2 vario value transmitted in m/s is used.

However, not all transmitters allow a setting of the sound output in order to hide certain areas or the sensitivity of the acoustic output.

Therefore, the UniLog 2 has the following setting options, to influence the transmission of the variometer in all telemetry systems:

- **"Vario Schwelle"** specifies the response threshold for the vario signal via telemetry separately for ascending and descending. A vario tone is generated by telemetry only if the rise / fall is greater than the threshold.
- **"Vario Ton"** determines whether the Vario is active when rising / falling or both. Here the Vario tone can also be switched off completely.
- **"Vario Faktor"** determines the factor by which the values of the Vario are multiplied for the telemetry. Normally 1.0 is set here, so that the real m/s is displayed on the telemetry. In particular cases, however, with a factor greater than 1, the sound output of the telemetry can be made more sensitive. If the transmitter itself does not allow such a setting.
- **"Vario Filter"** allows an adaptation of the response speed of the vario. "Slow" corresponds approximately to the previous evaluation, "medium" and "fast" are correspondingly faster. At the same time, however, the variable signal of the vario is always quicker.
- **"Vario bei Motor"** determines whether the Vario is active while the engine is running.

Example:

- "Vario Schwelle Steigen" (climb) is set to 0.5 m/s, "Vario Schwelle sinken" is set to -1.0 m/s
- "Vario Ton" is set to "Auf" (up)

→ if the model rises faster than 0.5 m / s, the value is transmitted

→ if the model rises more slowly or sinks, 0 is transmitted by the vario

If the Vario value is always to be transmitted, the "Vario Schwelle" must be set to 0.0 m / s and "Vario Ton" is set to "auf/ab" (up / down).

6. Connections of the UniLog 2

6.1. Top side

Connectors for external sensors A1, A2, and A3.

Here the temperature sensor (also PT1000), speed sensor, or other voltage signals are attached.

The three inputs have identical functions.

A voltage should never be supplied here! The UniLog 2 can be destroyed by this

Connector for rpm sensor.

The rpm sensor can be connected here. Our optical sensor, magnetic sensor, or brushless sensor are suitable. Propeller blades, gear factor, or brushless poles can be entered in the settings.

A voltage should never be supplied here! The UniLog 2 can be destroyed by this.

Telemetry connector "Link".

Here the telemetry is connected to the Rx over the supplied Patch cable. Cable and connector are identical for all telemetry systems.

The UniLog 2 can be supplied here with 3.8v to 10v

Connector for Receiver (Rx) and controller (ESC).

An Rx signal can be measured here. In Limiter mode the signal is looped through the UniLog 2 and affected accordingly.

The connection is galvanically separated from the UniLog 2

Memory card. Socket for the micro SD card. Both SD and SDHC cards can be used

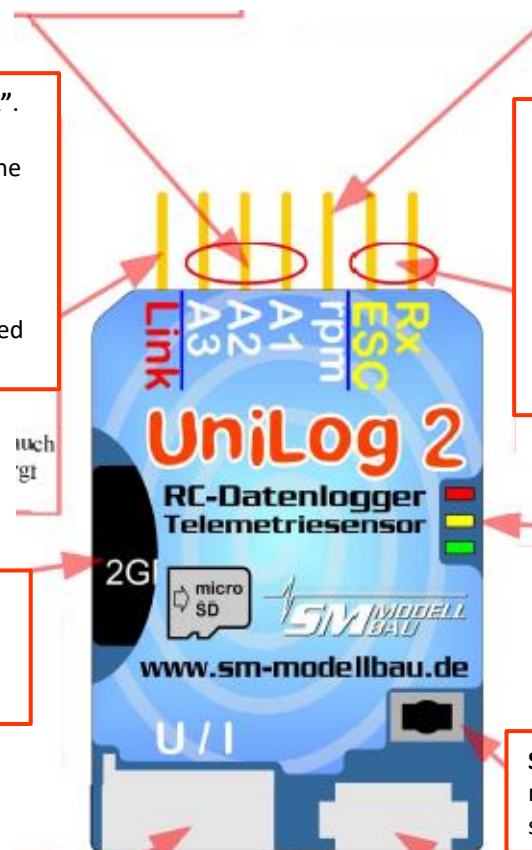
Signal LEDs.

Green shines: ready
Green flashing: recording
Red flashing: memory card missing

U/I socket. For the connection of the current and voltage sensor

Start/stop button. For manually starting and stopping the recording

COM socket. For the UniDisplay or PC, or the connection of the GPS logger



6.2. Under side



Balancer Connections

The balancer plug of a Lipo battery of up to 6s can be attached here to log the individual cell voltages.

It is important that the “Ground” or “–ve” connection is made to the right hand connector marked GND. The UnLog 2 is not powered over this connection. If only the cell voltages are being measured, it must also be connected to the receiver via the telemetry “Link” connection.

The Lipo battery must also have a connection to the supply voltage.

In the case of batteries with more than 6 cells the balancer plug of the battery which is connected to the minus pole of the regulator must be used.

6.3. Pin allocation on the pin row



The pins on the “pin row” are allocated as follows:

Top	Signal
Middle	Plus
Bottom	Minus

7. Connection examples

The UniLog 2 can be supplied with power either from the current sensor, or via the Telemetry connection “Link”, or from both at the same time.

7.1. Altitude only e.g. in the F3B/F3J Sailplaner or also HLG

For pure altitude it is only necessary to connect the UniLog 2 with the receiver lead provided. Like this the altitude logger weighs only 9 g plus leads. At the UniLog 2 the cable is attached to “Link”, and at the receiver either directly to the telemetry connection or, if no telemetry is present, with a free servo connection. In this way the UniLog 2 is supplied with power from the receiver.



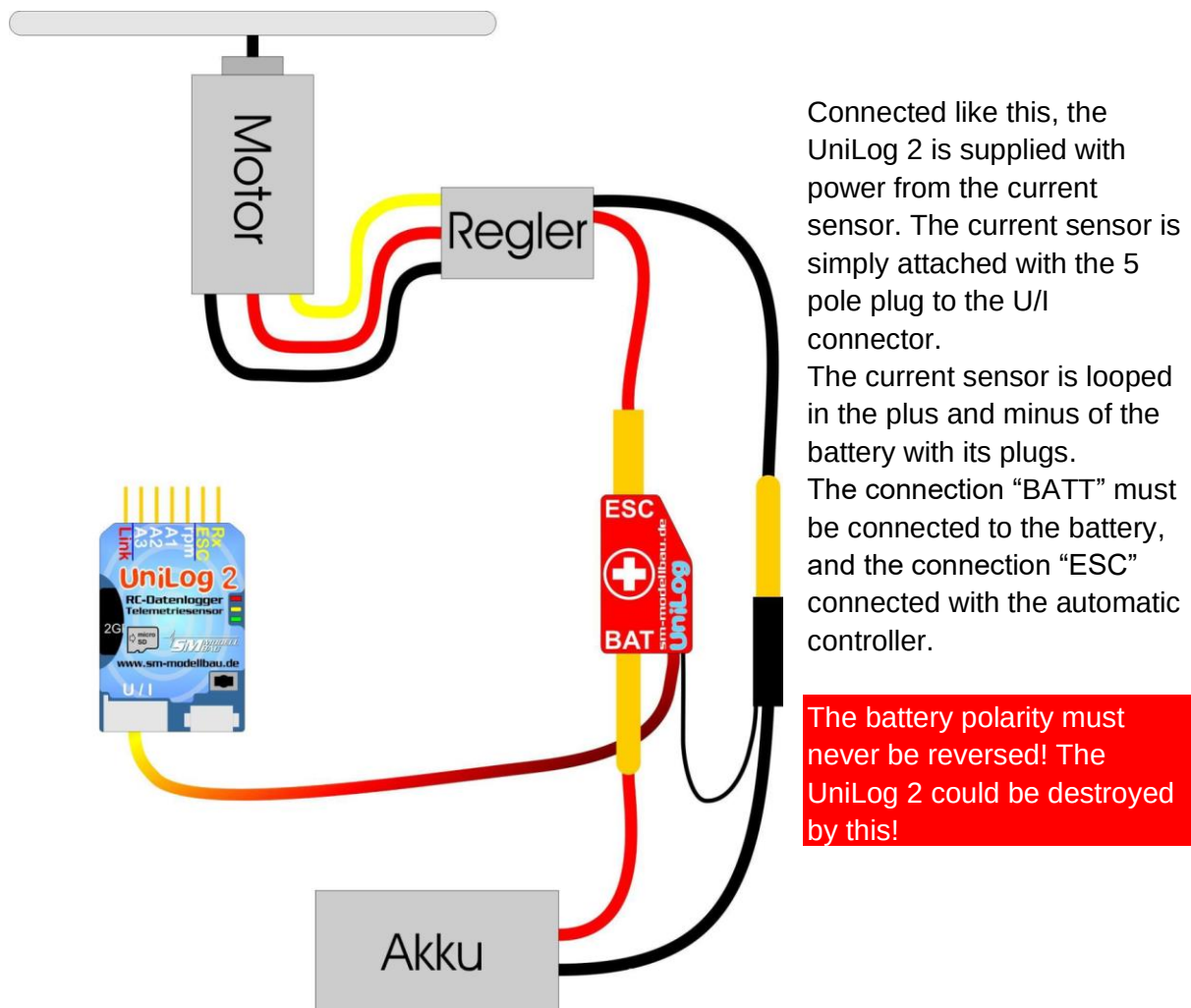
If the recording is to be started and stopped by the transmitter (with activated “Rx Start”) a second connection between the desired receiver channel and the Rx connector of the UniLog 2 is also necessary.

7.2. Electric drive measurement

For measurement of an electric drive a current and voltage sensor needs to be added to the UniLog 2. These sensors are available for different currents and with different plug systems:

Part number	plug system	max. Current	max. Voltage	Resistance	mass
2510/2530	2mm gold	-10A,+40A	60v	1 mOhm	7g
2512	MPX green	-10A,+80A	60v	1 mOhm	11g
2532	Deans Ultra	-10A,+80A	60v	1 mOhm	12g
2513/2533	4mm gold	-20A,+150A	60v	0.5 mOhm	14g
2514/2534	5.5 mm gold	-20A,+150A	60v	0.5 mOhm	15g
2515/2535	3.5 mm gold	-10A,+80A	60v	1 mOhm	11g
2516/2536	6 mm gold LMT	-20A,+150A	60v	0.5 mOhm	15g
2517/2537	6 mm gold LMT	-50A,+400A	60v	0.2 mOhm	18g
2523	Receiver current sensor GPN/FUT	-5A,+20A	10v	4 mOhm	9g

If a battery with more than 8s Lipo or 35 V is attached to the current sensor, the UniLog 2 must also be supplied via the telemetry connection "Link". Otherwise the voltage regulator in the current sensor is overheated and the UniLog 2 shut down. Thus with high drive voltages the UniLog 2 must be supplied with power by the current sensor AND receiver.

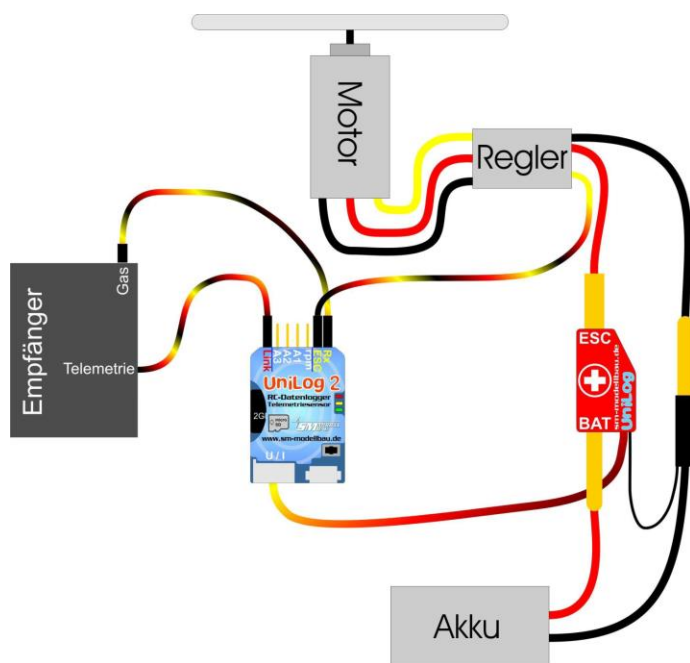


The current sensors for 150 and 400 A have usually no plug on the single negative conductor. A small connector plug should be fitted, with which the connection to the negative pole of the battery can be made. Here 2 mm gold plugs work well.

When switching the UniLog 2 on the current zero point is also calibrated. The current flowing at that time is thus set as 0 value.

7.3. Limiter for F1Q/F5B/F5D/F5F/F5J/F3G

With use as a Limiter in the competition classes F1Q/F5B/F5D/F5F/F3G the drive is switched off by the unilog 2 after consumption of the set energy quantity. For this, the signal from receiver to controller must be looped through the UniLog 2. The signal is read, recorded, and sent on again from the UniLog 2 “ESC” output..



Since the UniLog 2 intervenes actively in the servo signal, only the motor control channel should be sent through the UniLog 2.

In addition it must be ensured that the motor controller is OFF with a servo impulse of 900 us = 0.9 ms.

Connection for F5J:

Here no current sensor is needed. To power the UniLog 2 still make the connection from "Link" to the telemetry port of the receiver. When no telemetry is used, a connection for the supply between "Link" and a free servo channel should be made, but here you have to break the impulse line, only plus and minus are connected.

In the limiter operation, special attention is required if the receiving system is switched off sharply. In this case, it is technically possible that due to the shutdown the UniLog 2 can be supplied with an invalid servo pulse for the controller. Under certain circumstances, this can lead to a brief start-up of the engine!

7.3.1. Characteristics as Limiter:

- The limit is freely adjustable over PC, UniDisplay and telemetry
- The mode is selectable for:
 - F1Q - Signaling the timer component of the model over A2 and A3
 - F5D - Release of the engine again after 10 s
 - F5B/F - Final shut down
- The Limiter mode shows for 2 seconds after the run of LEDs when switching on
 - F1Q - the orange LED and the red LED
 - F5D - the orange LED and the green LED
 - F5B/F - the orange LED
 - F5J – the red LED
 - F3G – the red LED and the green LED
- The energy counter is set to 0 when switching on and cannot be reset later
- The Limiter is independent of the data capture
- When the limit is reached the red LED flashes fast, a servo impulse of 900 us is sent

- With Limiter set the memory rate is always switched to 20 Hz
- Power levels under 25 Watts (e.g. BEC) are ignored during the energy counting when a limit of more than 400 Wmin is set.

- In F5J mode the UniDisplay after connection automatically shows a new screen with the measured F5J values according to the current F5J rules. In the second row are the Firmware version and serial number. In the middle is the Starting height of the last flight. At the bottom for checking are some current (live) values, such as height, the timer for the 30 + 10 second interval, and the measured servo pulse in and out.



7.4. Connection of Analogue sensors at A1 to A3

On the UniLog 2 up to three analogue sensors can be attached at connector locations A1, A2, and A3, and be recorded.

All three connections are identical in function. The attached sensor needs only to be selected in the Setup of the UniLog 2. In this way up to three temperatures can be recorded at the same time, or two temperatures and speed with the speed sensor etc.

Settings options for A1, A2 and A3:

- Temperature sensor (part no. 2220 & 2221), measuring range of - 40 °C to + 125 °C
- Voltage in mV (measuring range is 0 to 3300 mV) for your own sensors
- Speed sensor to 250 km/h (stock no. 2560)
- Speed sensor to 450 km/h (stock no. 2560)
- Temperature sensor PT1000 (stock no. 2225), measuring range of - 50 °C to + 300 °C, with suitable PT1000 sensors also to + 800 °C

7.5. Min and Max values via receiver channel

Optionally, a free receiver channel can be used to switch between the transmission of Live, Minimum and Maximum values. Some telemetry systems can store the current maximum values in the transmitter. For systems that require do not have this option, you can use this option of UniLog 2 after the flight to switch to the minimum or maximum values in the telemetry with a simple switch on the transmitter. A second connection is needed between the desired free receiver channel and the "Rx" input of the UniLog 2.

At the transmitter a 3-way switch is programmed for this free channel which switches the channel between the following values:

- 100% for the minimum values (switching point <1.3 ms servo pulse width)
- 0% for the live values
- + 100% for the maximum values (switching point > 1.7 ms servo pulse width)

In addition, the option "Rx Eingang" must be set to "Min/Live/Max" in the UniLog 2 settings.

During operation, by means of a rapid 3 times toggle of the 3-way switch on the transmitter the minimum and maximum values can be reset.

If the capacity measurement is set to "immer weiter" (always continue), this reset of Min / Max also resets the capacity manually to 0. Prerequisite is that the model, after power on, was not yet in the air.

7.6. Connection of an rpm sensor

As a "number of revolutions" sensor, optical sensor part no. 2210, magnetic sensor part no. 2211 or Brushless rpm sensor part no. 2213 can be used at the "RPM" connector.

In the setup of the UniLog 2 for rpm measurement the correct number of measuring impulses per revolution must be given. With the optical and magnetic sensor that is the number of the propeller blades and/or magnets. With the Brushless rpm sensor that is the number of engine poles. A classical in-runner like e.g. a Lehner or a Hacker has 2 poles. An out-runner often has 10 or 14 poles. The data sheet for the motor should specify the number of poles.

In addition the gear ratio can be specified. So if an in-runner with gearbox is measured with the Brushless rpm sensor the gear ratio can be factored in. In this way the actual propeller rpm is recorded.

If the rpm is measured directly at the propeller and/or output (optical sensor at the propeller or magnetic sensor on the shaft), the gear ratio can be set at 1,0:1.

Connection of the Brushless rpm sensor (part no. 2213):

The sensor is attached via the soldered on 2 mm gold plug to one of the three motor wires. Thus a connection to one of the three wires between the automatic controller and motor must be made. This can be for example the included 2 mm socket soldered directly with the ESC cable, or tapped in with a thin cable (no current flows here).



Brushless rpm measurement functions only if the UniLog 2 also has a bonding to the drive at the same time via the current sensor.

7.7. rpm measurement for gasoline engines directly from the ignition

As of firmware v1.12, the UniLog 2 offers the possibility to directly convert the digital speed signal of an electronic Ignition. Many newer ignition systems have a separate output for a speed indication that gives an impulse at each ignition. This signal can be evaluated and recorded by UniLog 2, and of course transmitted by telemetry.

However, in this case, the galvanic separation of the two circuits from the ignition and the reception system is not possible. The UniLog 2 is bridged. Therefore the internally galvanically decoupled "Rx" connection is used instead of the connection "rpm".

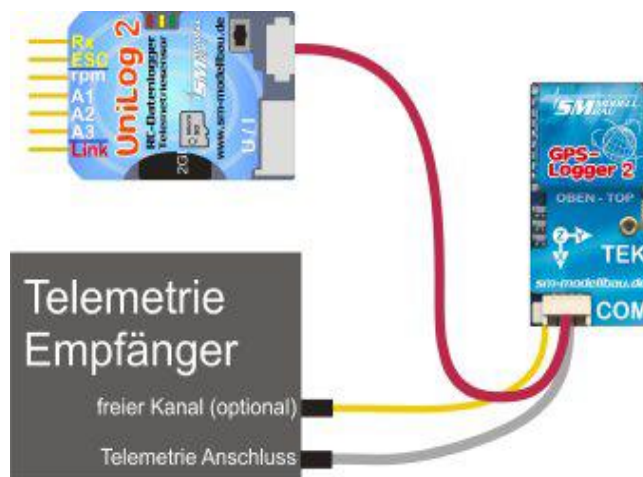
With the option "Rx Eingang" set to "Drehzahl", the input "Rx" becomes the rpm input. The rpm input is then Automatically Rx input, the inputs are exchanged internally

The signal of the ignition must be fed into the "Rx" connection of the UniLog 2, including the supply voltage. It must therefore have a three-pole connection with minus, plus and the signal from the Ignition.

Please do not place the signal of an ignition at the input "rpm"! Galvanic isolation is missing and In addition, no external voltage can be fed in here.

Some ignitions have no separate cable for the speed signal but the pulse line of the supply connection is used for this purpose. This is the case e.g. with the new Valach ignition with the plastic housing. Here it is sufficient if a Y-cable before the ignition is switched and one end goes to the battery (only Plus and minus) and the other with plus minus and signal to UniLog 2.

7.8. Connection to the GPS Logger



With the leads part no. 2720 or 2721 the UniLog 2 can be directly connected with the GPS Logger. The UniLog 2 is supplied with power via its current sensor, while the GPS Logger is connected directly to the receiver. In this way the GPS Logger also records automatically the data from the UniLog 2 on its memory card. Thus GPS data and the recorded values on the UniLog 2 are synchronised and can be evaluated together. The recording on the GPS Logger takes place with the GPS Logger memory rate i.e. 10 Hz max.

For Jeti Duplex, some telemetry data from the UniLog 2 is passed on to the receiver by the GPS-Logger. This can save the otherwise necessary expander E4 from Jeti. The UniLog 2 is supplied with power via the "Link" connection at a free receiver slot, While the GPS logger is connected to the receiver's telemetry port. See also the corresponding point in the manual of the GPS-Logger 2.

The connection of the UniLog 1/2 or UniSens-E only works with HoTT, Jeti and M-Link operation of the GPS-Logger 2. The COM interface cannot be used with the other telemetries.

Note: Only the three-core cable connections part no. 2720 and 2721 may be used! With the 4 core cable part no. 2401 the two internal voltages are connected which can lead to faults.

8. Use of the UniDisplay

For connection of the UniLog 2 firmware v1.26 or higher must be used in the UniDisplay. An update for the UniDisplay can be downloaded free of charge from our homepage (www.sm-modellbau.de).

UniDisplay and UniLog 2 are connected with the cable provided with the display. The connector location is labelled “COM” on the UniLog 2. The cable connection can be made either way round, which end is used is irrelevant.

The display is powered by the UniLog 2 and turns on automatically as soon as the UniLog 2 is connected. The display can be attached at any time to the UniLog 2.



Menu:

First the menu is activated. The menu options can be selected with the “plus” and “Minus” buttons, and the appropriate item selected with “Enter”

Live data display screen 1:

Here all current readings are displayed. Most values are self-explanatory.

- “Plus” starts and stops the recording.
- “Minus” changes between Live/MIN/MAX values.
- “Enter” changes between the Live screens 1, 2 and 3.
- “Esc” changes back to the menu.

Top right is the current file number. Including changes in the past, time, date and time.

The last line shows the servo impulse measured at the Rx connection and then at the ESC output connection.



Live data display screen 2:

On the second Live display screen are shown the 6 single cell voltages and the values of the ports A1 to A3.



Live data display screen 3:

Here are shown the current air pressure measured by the barometric pressure sensor, and the internal temperature of the UniLog 2. Because of self-heating this temperature is always a little higher than the ambient temperature.



Data display from the memory:

In the first screen the desired file is selected from the memory card. Scroll through the files with “plus” and “Minus”, and select the file with “Enter”.



The memory display is then almost identical to the Live Values: “Plus” and “Minus” to scroll through the readings every single value can be displayed. A long press accelerates the run. If available, the MAX and MIN values are displayed at the end of the file. A long press on “Enter” scrolls directly to the end of the file.



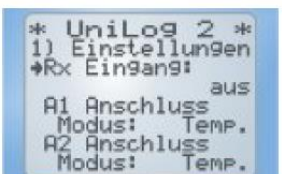
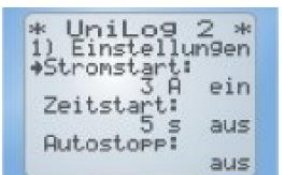
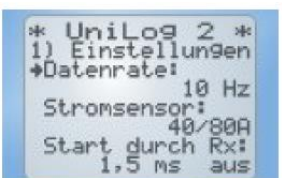
Setup:

Here the menu for all settings of the UniLog 2 appears. In the second line is the firmware version of the UniLog 2 and also the serial number. Move through the menu options with “Plus” and “Minus”, and select the appropriate point with “Enter”.



Settings screens

Here the settings of the UniLog 2 are summarized. Move through the menu options with “Plus” and “Minus”, and select the appropriate point with “Enter”. The Arrow then becomes a Dot and the selected value can be changed with “Plus” and “Minus”. A press on “Esc” or “Enter” stores the change.



- **“Telemetrie”**: setting for the telemetry used
- **“HoTT Modus”**: desired mode with HoTT
- **“Kapaz.Messung”**: enables or disables the continued capacity measurement
- **“Stromoffset”**: automatic zero point calibration on or off at startup
- **“Datenrate”**: selects the recording speed
- **“Stromsensor”**: selects the current sensor connected
- **“Start durch Rx”**: activates the recording start by receiver
- **“Stromstart”**: activates the recording start by motor current
- **“Zeitstart”**: activates the recording start after set time
- **“Autostopp”**: allows makes the automatic stop of the recording
- **“Min/Max Anzeige per Rx”**: allows the selection between Live/max/min values for display via the telemetry channel to a receiver
- **“A Anschluss modus”**: selects the sensor connected to these analogue ports
- **“Propeller”**: specifies the number of prop blades with optical and/or the number of fixed magnets with magnetic rpm measurement
- **“Pole”**: specifies the number of motor poles for Brushless rpm measurement
- **“Getriebefaktor”**: is the gear ratio for rpm measurement

```
* UniLog 2 *
1) Einstellungen
→A3 Anschluss
  Modus: Temp
  Propeller/Pole:
    2 / 4
  Getriebefaktor:
    1 : 1.00
```

```
* UniLog 2 *
1) Einstellungen
→Limiter-Modus:
  Limit Wert: aus
    1000 Wmin
  Varioton:
  Steigen/Sinken
```

```
* UniLog 2 *
1) Einstellungen
→Vario bei Motor
  Vario Steigen:
    ab +0.0 m/s
  Vario Sinken:
    ab -0.0 m/s
```

```
* UniLog 2 *
1) Einstellungen
→Vario Faktor:
  Vario Filter:
    mittel
  Uhrzeit:
    12:00
```

```
* UniLog 2 *
1) Einstellungen
→Datum:
  fixe Seriennr.:
    ja
```

- **“Limiter modus”** : defines the competition class for the Limiter
- **“Limit Wert”** : is the corresponding energy limit
- **“Varioton”** : specifies whether the Vario is active during climbing/sinking
- **“Vario bei motor”**: switch off the vario tone while the engine is running
- **“Vario Steigen”** : indicates the positive threshold for the Vario signal
- **“Vario Sinken”** : indicates the negative threshold for the Vario signal
- **“Vario Faktor”** : Determines by which factor the values of the vario telemetry can be multiplied.
- **“Vario Filter”** : Allows the response speed of the varios to be adjusted
- **“Uhrzeit”** and **“Datum”** are used for adjusting the internal clock
- **“fixe Seriennummer”** See Jei and Futaba, allows the exchange of sensors On different models without having to redo all the settings.

UniDisplay
sm-modellbau.de

```
* UniLog 2 *
2) Alarmer
→Strom 50.0A -
Startsp 12.3V +
Spann. 10.0V -
Kapaz 1800mAh +
Hoehe 200m -
RxSpann 4.70V -
```

esc — + ent

Alarms

The alarms that can be set here apply to all telemetry versions.

Depending on the telemetry, alarms are also set directly in the transmitter. Please note the instructions for the individual telemetry systems.

When the arrow on the left and the appropriate menu option activated with “Enter”, the value of the alarm can be changed.

If the arrow is moved right with “Plus” or “Minus” and the “Enter” key pressed to activate the menu point, the alarm can be activated (“+”) or deactivated (“-“) with “Plus” or “Minus”.

UniDisplay
sm-modellbau.de

```
* UniLog 2 *
3) M-Link Adress
→Spannung 02
Strom 03
Drehzahl --
Kapazitaet 04
Vario --
Hoehe 07
```

esc — + ent

M-Link addresses:

For transfer of Unilog 2 measured values by M-Link, addresses can be assigned here for the display on Multiplex transmitter. Each address may only be assigned once to any attached M-Link sensor, including the M-Link receiver. The Bus system ceases to function with multiple assignments. If a value is not to be transmitted chose the address “—“. This value becomes the highest permissible Address 15.



Jeti EX values:

Here, all measured values can be transferred to the Jeti transmitter or the JetiBox Profi and can be individually selected. Fewer measured values mean faster transmission, so you can hide the unneeded values here.



Spektrum Init:

The address specifies the assignment in the Spektrum adapter. Here is only important that with Several SM telemetry sensors on the Spektrum adapter no address is assigned twice. Normally the default is OK. This includes all Spektrum sensors supported and operated by UniLog 2 which can be individually selected or deselected. Each type of sensor may only occur once.



Futaba Init:

The compatibility mode for the Robbe Telemetry Box can be activated here.

9. Telemetry Operation

With the UniLog 2, the telemetry of Jeti Duplex (EX), Multiplex M-Link, Graupner HoTT, Futaba FASSTest S.BUS2, JR DMSS, frSky and Spektrum is supported.

The telemetry operation is similar for all applicable remote control systems: live data is displayed at the transmitter or an external display, with Jeti Duplex and Hott, the UniLog 2 can also be operated from the transmitter. If the system has a voice, then this is also supported by the UniLog 2.

The alarm output depends on the telemetry. In some systems, the alarm is generated by the UniLog2, in others thresholds are set directly on the transmitter. Please follow the instructions below.

9.1. Jeti duplex

The UniLog 2 is a complete telemetry sensor for Jeti Duplex 2.4 GHz systems. All measured values can be transferred live to the ground and displayed on the JetiBox. The Jeti Expander E4 for the connection of up to 4 sensors is supported.

Connection direct to the Jeti Duplex receiver is made with the patch cable supplied between "Link" on the UniLog 2 and "ext" on the Jeti Duplex receiver. With the REX receivers it must be ensured that the telemetry slot is configured to "JETIBOX / Sensor" or for a Central Box "Telemetry Input". This setting can be from the transmitter in the Device Explorer.

9.1.1. EX telemetry with the Jeti transmitters and the JetiBox Profi

The UniLog 2 also transmits data via EX telemetry from Jeti to the transmitters DC/DS-12/14/16/24 and to JetiBox Profi. On the transmitter, the display contents, voice output and alarms can be defined as desired. Using the built-in emulation of the JetiBox display, the UniLog 2 can also be operated from the transmitter and set as described below.

If a UniLog 2 is exchanged for another for the telemetry values in the DC / DS-14/16/24 or the Profi Box, each UniLog 2 has its own serial number and the Jeti EX system differentiates individual devices with it.

From firmware v1.12, however, the option "fixed serial number" can be activated. Every UniLog 2 then transmits serial number 12456 and the devices can be interchanged.

The following values are displayed in Jeti EX mode. The values are individually selectable via the PC program Or the UniDisplay. The Jeti transmitters can receive a maximum of 32 values at the same time. When using with expander or to increase the transmission rate, unimportant values can be blanked.

- | | | |
|--------------|-------------------|-------------------------|
| • Voltage | • Energy | • A1 |
| • current | • Power | • A2 |
| • capacity | • Air pressure | • A3 |
| • Rx voltage | • Pulse in | • Temperature internal |
| • Height | • Pulse out | • Height gain |
| • Vario | • Cell min | • Residual capacity in% |
| • rpm | • Cell min number | |

9.1.2. Operation of the UniLog 2 with the JetiBox

< - Mx
v

After the start of transmission the JetiBox is changed to Mx for the attached sensors.

Firmware v1.11
SM UniLog 2

A press the on the ▼ key changes to the UniLog 2 initialisation screen then the measured data is displayed.

A 23.28 V 221.8 m
36.04 A 1377 mAh

As soon as the first screen with measured data appears, the different data screens can be selected with presses of ◀ and ▶ keys. A pressure on the key ▲ starts the recording of data in the UniLog 2, which is indicated by an acoustic signal. A further pressure on ▲ terminates the recording.

A simultaneous long pressure on the keys ◀ and ▶ changes between the display of Live / MAX / MIN values.

In the top left of the screen there is an indicator of the current active data screen and/or the status of the UniLog 2:

- **A** first data screen, the following screens have B, C, etc.
- ***** recording running
- **>** maximum values are indicated
- **<** minimum values are indicated

< Current Alarm >
(OFF) 100A

A press of key ▼ changes to the settings. Again with the keys ◀ and ▶ the different screens and the desired point are selected.

< Current Alarm >
(ON) < 49A >

After a further press of key ▼ the selected value can then be changed (keys ◀ and ▶). With a simultaneous pressure on ▲ and ▼ the alarm is switched on/off.

Changed settings are only stored with the move back to the selection level with ▲.

9.1.3. Display of measured values on the JetiBox

A 23.28 V 221.8 m
36.04 A 1377 mAh

Top: Drive voltage, barometric altitude from starting point

Bottom: Drive current, capacity used

B 1750.1 Wmin
2481 rpm 839 W

Top: energy used

Bottom: rpm, drive power

C	5.01 VRx	221.8 m
	>>>>> _	+12.1 m

Top: Rx voltage, barometric altitude from starting point
Bottom: Graphical display of the Vario, Vario as numerical value

D	3.61	3.65	3.66
	3.65	0.00	0.00

Top: Single cells 1 - 3
Bottom: Single cells 4 - 6

E	A1	- - - - °C
	A2	44.9 °C

Top: Sensor value at port A1
Bottom: Sensor value at port A2

F	A3	221.9 Km / h
	1100 us	-> 1100 us

Top: Sensor value at port A3
Bottom: Servo impulse at the Rx connection, servo impulse at the ESC connection

G	00:14:34.5
	01.06.2011 14:55

Top: Recording time
Bottom: current date alternating with the file number, current time

H	971.43 hPa
	internal 28.1 °C

Top: current air pressure
Bottom: internal temperature of the UniLog 2.
As a result of self-heating this temperature is always somewhat higher than the ambient temperature.

9.1.4. Alarms

When operating on the Jeti transmitter modules, with the display of data on the simple JetiBox, all alarms and also the Vario sounds are generated directly from UniLog 2
All settings are also made on the UniLog 2.

The Jetibox Profi and the Jeti transmitter can, in Jeti EX mode, generate alarms and Vario tones themselves. These are then preset in the Box or transmitter. Alarms that are set in UniLog 2 remain as additional output.

9.2. Multiplex M-Link

The UniLog 2 is also a full telemetry sensor for the Multiplex M-Link 2.4 GHz system. The measured values can be transferred live to the ground and be displayed directly on the Multiplex the Royal Pro or COCKPIT SX transmitter.

In order to display the correct rpm, the Royal Pro transmitter must have as a minimum firmware V3.46, and the external display at least v1.09.

The connection to the M-Link receiver is made with the Patch cable provided between the connection location “Link” on the UniLog 2 and “Sensor” on the M-Link receiver.



The settings for telemetry can be made either with the UniDisplay (also see chapter 8) or with our “SM UniLog 2 tools” software on a PC.

The addresses for display on the Multiplex remote control (the line that the respective value is indicated in) can also be freely selected.

9.2.1. alarms

All alarms on M-link are produced directly from UniLog 2. All settings are also reset in the UniLog 2.

As an additional feature the UniLog 2 writes all data continuously on the Multiplex M-Link bus system and stores it in parallel with its own data on the memory card. So you can expand your M-Link sensor system to a practically infinitely large Data logger!

9.3. Graupner HoTT

The UniLog 2 is also a full telemetry sensor for the Graupner HoTT 2.4 GHz system. The measured values can be transmitted live to the ground and displayed directly on either the Smart Box at the HoTT transmitter or directly in the display of the HoTT transmitter. The connection to the HoTT receiver is made with the Patch cable provided, between the connection "Link" on the UniLog 2 and "T" on the HoTT receiver.

The UniLog 2 can be operated with HoTT as a "General Module (GAM)", "Electric Air Module (EAM)", "Controller (ESC)" or as "Vario". This also means that several UniLog 2 or combinations with the UniSens-E or other sensors can be used.

9.3.1. Alarms

The UniLog 2 supports both the Text Mode" and the "Digital Mode" of the HoTT system. In both modes of operation all adjustable alarms on the UniLog 2 are indicated at the transmitter by beeps or speech output.

9.3.2. Vario

The HoTT transmitters display the range from -0.5 m / s to 0 m / s in the sound output. However, to obtain a continuous Vario tone, the UniLog 2 skips this area. For a more sensitive response of the acoustic output HoTT should have a "vario factor" of 2 to 4 set in the UniLog 2. The result is a good resolution acoustic output without a hole around the zero point. You have to accept, however, that the Vario value displayed and recorded in the transmitter does not correspond to reality.

9.3.3. Text Mode

For text mode you get there via Telemetry menu and "Settings view". The correct mode can be set using the "up" and "down" on left panel of the transmitter according to the HoTT mode set in the UniLog 2 .With one click, to the right go from the receiver data to the text display of the UniLog 2



This operation is done with the right touch pad on the transmitter.

Structure and contents are completely identical to the screens of the UniDisplay, (see chapter 8 for the UniDisplay). Here all alarms can be set, then indicated at the transmitter via beeps or speech output.

9.3.4. Digital Mode

From the default display of the transmitter using the "left" and "right" of the left touchpad digital mode is activated. The correct mode can be set using the "up" and "down" on left panel of the transmitter according to the HoTT mode set in the UniLog 2.

Use the "left" and "Right" of the left touchpad to switch between the different screens. Depending on the mode there are more screens with great display of measured values.

Display as GAM (General module):

Some values from the UniLog 2 are shown differently:

Fuel scale Shows the remaining battery capacity in accordance with the set alarm

Temperature 1 shows the temperature of a sensor attached to A1-A3

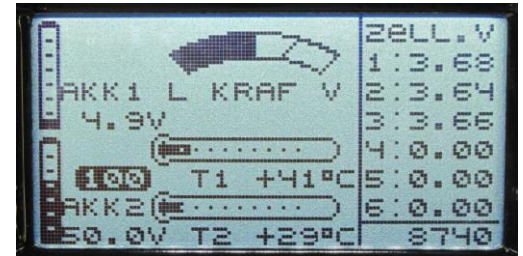
Temperature 2 shows the internal temperature or temperature to A2 – A3 if at least 2 temperature sensors are connected

Battery 1 shows the Receiver voltage

Battery 2 indicates whether, by Rx control, minimum or maximum values are selected.

50.0V → indicates live values, 0.0V → minimum values, 99.9V → maximum values

m3 shows the height gain of the last 10 seconds.



Display as EAM (Electric Air Module):

These values are configured differently from UniLog 2:

cell voltages: cell 1 to 6 show as L and H the same values, Cell 7 is free.

Fuel scale

Shows the remaining battery capacity in accordance with the set with alarm

Temperature 1 shows the temperature of a sensor attached to A1-A3

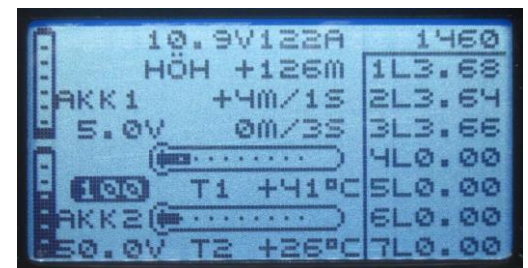
Temperature 2 shows the internal temperature or temperature to A2 – A3 if at least 2 temperature sensors are connected

Battery 1 shows the Receiver voltage

Battery 2 indicates whether, by Rx control, minimum or maximum values are selected.

50.0V → indicates live values, 0.0V → minimum values, 99.9V → maximum values

m3 shows the height gain of the last 10 seconds.

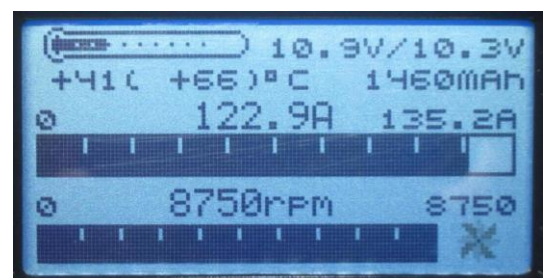


Display as ESC (Controller):

These values are configured differently from UniLog 2:

Temperature: the temperature fields show the four In turn, the temperature sensors A1, A2, A3 (if available) and the internal temperature.

In this mode, cell voltages, altitude and Vario cannot be displayed.



9.4. Futaba S.Bus2

The UniLog 2 is connected to the S.BUS2 slot of the receiver just like any other sensor

Currently the UniLog 2 is not yet registered in the transmitters, so it uses already existing sensor protocols.

We tested the integration with the T10J with the R3008SB receiver, and the T14SG, the FX-32, the T18SZ and the T18MZ with the R7008SB and R7003SB receivers. Always use the latest firmware version the transmitter, because older versions may not completely support the integration.

With S.BUS2 Servo data sensor values can be connected to the same data line. But since the servo data is far more important than the sensor values we strongly recommend that you make a strict separation. All servos go to the S.BUS1 connection of the receiver, all the sensors on the S.BUS2. Thus, in the event of an error, a sensor can never interfere with the data for the servos.

If nevertheless the UniLog 2 is to be operated together with the servos on S.BUS2, is absolutely mandatory that a connection cable No. 9110 is used between UniLog 2 and S.BUS2! Thus the sensor is decoupled from the bus so far that any influence on the servo data is impossible.

9.4.1. Registration on the Transmitter

To use the UniLog 2 with the S.BUS2, it must like all S.BUS2 sensors be first registered on the transmitter. For this purpose the "Link" connection of the UniLog 2 is connected to the "S.I / F" socket on the transmitter. Some transmitters, such as the T14SG, do not provide power on this jack, in this case the UniLog 2 can be supplied with a separate battery via a Y cable. The UniLog 2 behaves like a Robbe / Futaba sensor and is integrated into the system in the same way. Please also refer to the transmitter instructions.

However, the UniLog 2 currently uses ten Robbe / Futaba sensor values to represent all measured values. Some values cannot be mapped to the correct unit. Here one must remember the assignment with the T14SG, in the T18MZ you can rename the sensors and thus the assignment is easier to understand.

Since the sensor F1678 can only represent currents up to 160 A the UniLog 2 with a 400 A sensor connected sends the current sensor values / 10 , ie 400 A measured value will be shown on the display as 40.0 A .

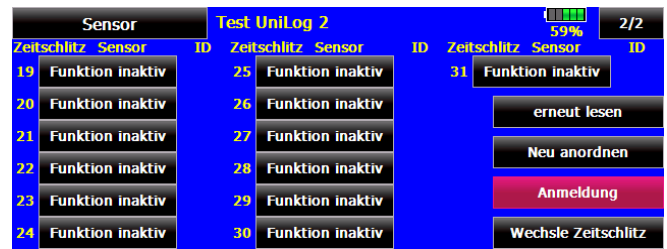
As of firmware v1.13, the UniLog 2 sends the reception quality data output from the receiver S.BUS2 back to the transmitter as a further temperature sensor TEMP125. This allows this Value to be used for corresponding alarms in the transmitter. % Values are output in the steps 25/50/75/100.

9.4.2. Robbe Telemetry Box

The Robbe Telemetry Box as a further possibility of the presentation of the Futaba telemetry data unfortunately does not recognise the temperature sensor SBS-01T used by UniLog 2. Therefore there is the option "Telemetry Box" In the settings of UniLog 2, which registers all Futaba SBS-01T sensors through the sensor TEMP125. Thus the UniLog 2 can also be used with the Telemetry Box without major restrictions.

The example of the T18MZ here follows the steps of the application:

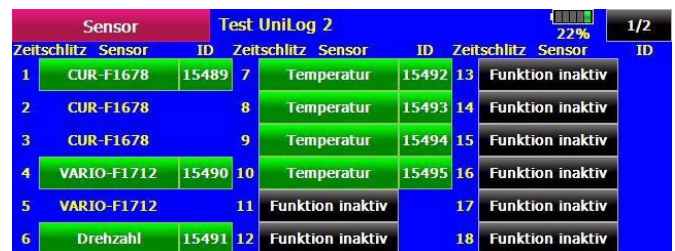
If the UniLog 2 connected to the transmitter and is supplied with power, it is invoked in the sensor menu item "Login". In this way the sensor is registered in the transmitter and assigned free slots. The sensor and transmitter save this assignment.



To be able to represent all values the menu item "Login" on the transmitter must necessarily be called ten times. The registration is terminated when the message "sensor already exists" appears.

When the application of all sensors is complete, the sensor list looks like this:

In T18MZ the sensors can then be renamed.



13 slots are occupied by the nine UniLog 2 sensors:

Sensor	Name	Slots	Original designation	In Unilog 2	Value examples
1	CUR-F1678 3	3	Current Volts Capacity	Strom Spannung Kapazität	141.7 A 37.1 V 2345 mAh
2	VARIO-1712	2	Height Vario	Hohe Vario	258 m 83.2 m/s
3	SBS 01RM / O	1	U/min	Drehzahl	8736 rpm
4	SBS-01T	1	Temperature	lowest cell voltage alternating with the cell number	3697 °C for 3,697 V 1 °C = cell 1
5	SBS-01T or SPEED-1677	1	Temperature or SPEED	Sensor A1 or Speed sensor A1	312 °C for °C or mV, 312, km/h
6	SBS-01T or SPEED-1677	1	Temperature or SPEED	Sensor A2 or Speed sensor A2	312 °C for °C or mV, 312, km/h
7	SBS-01T or SPEED-1677	1	Temperature or SPEED	Sensor A3 or Speed sensor A3	312 °C for °C or mV, 312, km/h
8	SBS-01T	1	Temperature	Energy	2345 Wmin
9	TEMP125	1	Temperature	Residual capacity as a value	0 - 100 (%)
10	TEMP125	1	Temperature	Reception quality from the receiver data	Steps 25/50/75/100

Now connect the UniLog 2 to the receiver and call up the transmitter telemetry display.

Here again the T18MZ for example.

Telemetrie		Test UniLog 2		Y.II	24%	1/2
Empfänger	Extern	UniLog 2(Kapazität)	6. Drehzahl(Magnet.)			
5.1V	0.0V	2345mAh	8736rpm			
UniLog 2(Strom)		UniLog 2(Höhe)	min. Zelle			
141.7A		258m	3697°C			
UniLog 2(Spannung)		UniLog 2(Vario)	A1 Speed			
37.1V		83.2m/s	312°C			

Telemetrie		Test UniLog 2		Y.II	24%	2/2
A2 Temp. Akku		41°C				
A3 Temperatur		0°C				

See the same values in the T14SG as follows (here the names cannot be changed):

TELEMETRIE		Y.II	1/3
Rx-BATT.	1 STROM		
5.1V	+141.6A		
EMPFÄNGR	CURR-1678		
EXT-VOLT	1 SPANNUNG		
0.0V	37.1V		
EMPFÄNGR	CURR-1678		

TELEMETRIE		Y.II	2/3
1 KAPAZIT.	4 VARIO		
+2,345mAh	+83.2m/s		
CURR-1678	VARIO-1712		
4 HÖHE	6 U/min		
+258 m	8,736rpm		
VARIO-1712	SBS-01RM/O		

TELEMETRIE		Y.II	3/3
7 TEMPERAT	9 TEMPERAT		
+3697°C	+41°C		
SBS-01T	SBS-01T		
8 TEMPERAT	10 TEMPERAT		
+312°C	+0°C		
SBS-01T	SBS-01T		

9.4. 2 Alarms

Since the UniLog 2 uses existing sensors for the display, there are a few features in the setting of alarms. In principle with S.BUS2 the alarms defined in the transmitter. The UniLog 2 has no way to directly activate an alarm at the transmitter.

Required settings:

- Alarm for altitude and vario → set inside the transmitter on HEIGHT and VARIO
- Alarm for current → set inside the transmitter on CURRENT
- Alarm for voltage → set inside the transmitter on VOLTAGE
- Alarm for starting voltage → set inside the transmitter on VOLTAGE and additional alarm. Enable start voltage in UniLog 2 and specify the desired value there → the UniLog 2 then transmits in case of alarm a constant from 0 to 50 volts continuous voltage value at VOLTAGE and gives the alarm.
- Alarm for capacity → two options:
1. only set inside the transmitter on CAPACITY
 2. in the transmitter at CAPACITY set an alarm for decreasing values of less than 0 (alarm with down arrow) and enable an additional capacity alarm in the UniLog 2 and specify the desired value there → the UniLog 2 then transmits a negative current value to trigger an alarm
Advantage: the UniLog 2 can thus affect the alarm itself and deactivate automatically after 20 seconds and also at the landing.
- Alarm for individual cell voltage → in the transmitter at TEMPERATURE # 1 set an alarm for values less than 0 ° C. (Alarm with down arrow) and enable an additional cell voltage alarm in the UniLog 2 and specify the desired value there → the UniLog 2 then transmits -100 ° C to trigger an alarm and alternates with the current measured value.
- Alarm for A1, A2 and A3 → in the transmitter at TEMPERAT # 2/3/4 set an alarm value lower than 0 ° C (Alarm with down arrow) and enable an additional A1/A2/A3 alarm in the UniLog 2 and specify the desired value there → the UniLog 2 then transfers -100 ° C to trigger an alarm and the display alternates with the actual measured value.

9.5. JR Propo DMSS

The UniLog 2 is connected to the sensor slot of the receiver like any other sensor and the following data is transmitted:

- rpm (sensor address 0x02 "rotation")
- air pressure, altitude, vario (sensor address 0x03 "Pressure / Altitude")
- Air speed (only with speed sensor) (sensor address 0x05 "Airspeed")
- voltage, current, capacity, watts,
Single cells 1-6 (sensor address 0x08 "Power Pack")

In operation as limiter F5B / F / D, the value for the energy is transferred in Wmin instead of the power.

No more sensors occupying the same addresses can be connected. For the free addresses, further sensors can easily be plugged into the receiver in parallel to UniLog 2 with a Y cable. The connection was tested with the XG8 transmitter and the RG831B and RG731BX receivers. Always use the latest firmware version on the transmitter, because older versions may not fully support the integration.

9.5.1. Presentation on the Transmitter



# 1	1:24	6.4V
HÖHE	F-AKKU U	
+155.2m	20.35V	
VARIO	F-AKKU A	
+13.2m/s	48.42A	
UPM	F-AKKU C	
9266UPm	1388mAh	

The values can be displayed directly and the sequence on the display is freely selectable.

Special feature: The transmitter converts the battery capacity into a residual capacity. The measured value of the UniLog 2 is therefore deducted from the default setting in the transmitter. In the example shown, there are still 1388 mAh in the battery.

9.5.2. Alarms

The alarms are defined in the transmitter in principle with JR DMSS. The UniLog 2 has no way to directly trigger an alarm on the transmitter. All alarm thresholds, and also the production of the Vario tone, are specified in the transmitter.

Required settings:

Alarm for current → is not currently supported

Alarm for starting voltage → voltage alarm set in the transmitter and in addition a Start Voltage alarm enabled in the UniLog 2 and the desired value specified there → the transmitter then displays in case of alarm a voltage value of 1:00 V and gives the alarm.

Alarm for capacity → Since the transmitter calculates the remaining battery capacity from the measured value of the UniLog 2, the actual battery size and the desired percent residual capacity is simply determined in the transmitter.

9.6. FrSky

With the FrSky telemetry all receivers with the new "S.Port" connection are supported. For telemetry operation the receiver must be connected in the D16 mode!

The UniLog 2 is connected to the sensor slot of the receiver like any other sensor and Transfers the following data:

• Current	"Curr"	
• Capacity	"A4"	units must be changed to "mAh" in the transmitter
• Capacity		as the value "Consumption" calculated from the current "Curr"
• Voltage	"VFAS"	
• rpm	"RPM"	must be set to 1 sheet in the transmitter
• Altitude	"Alt"	
• Vario	VSpd	Units "m/s"
• Residual Capacity	"Fuel%"	is calculated from the capacity consumed and the alarm threshold in the UniLog 2
• Energy Wmin	"Fuel ml"	
• A1, A2, A3	"Tmp1"	when connecting a temperature sensor
	"ASpd"	when connecting a speed sensor
	"A3"	when measuring an analog sensor in mV (unit "V" instead of "mV")
• Cells	"Cels"	with indication of the individual voltages, Min, Max and difference as calculated values

Additional values can be calculated in the transmitter. Power e.g. by multiplication of Voltage (VFAS) and current (Curr).

The sensor address (ID) of the UniLog 2 in the FrSky system can be freely selected in the settings. No other sensors occupying the same addresses can be connected. For free addresses further sensors are simply plugged in with a V cable parallel to the UniLog 2 on the receiver.

9.6.1. Presentation on the Transmitter

UniLog 2 8.2V Tmp1 00:00			
Alt	258.2	Curr	141.7
VSpd	83.2	VFAS	37.15
Leis	526.4	Ener	2345
FSpd	112.6	Cmin	3.69
		Cdiff	0.03

All values can be displayed directly. The sequence on the display is freely selectable. The energy was renamed here from "Fuel" to "Ener".

9.6.2. Alarms

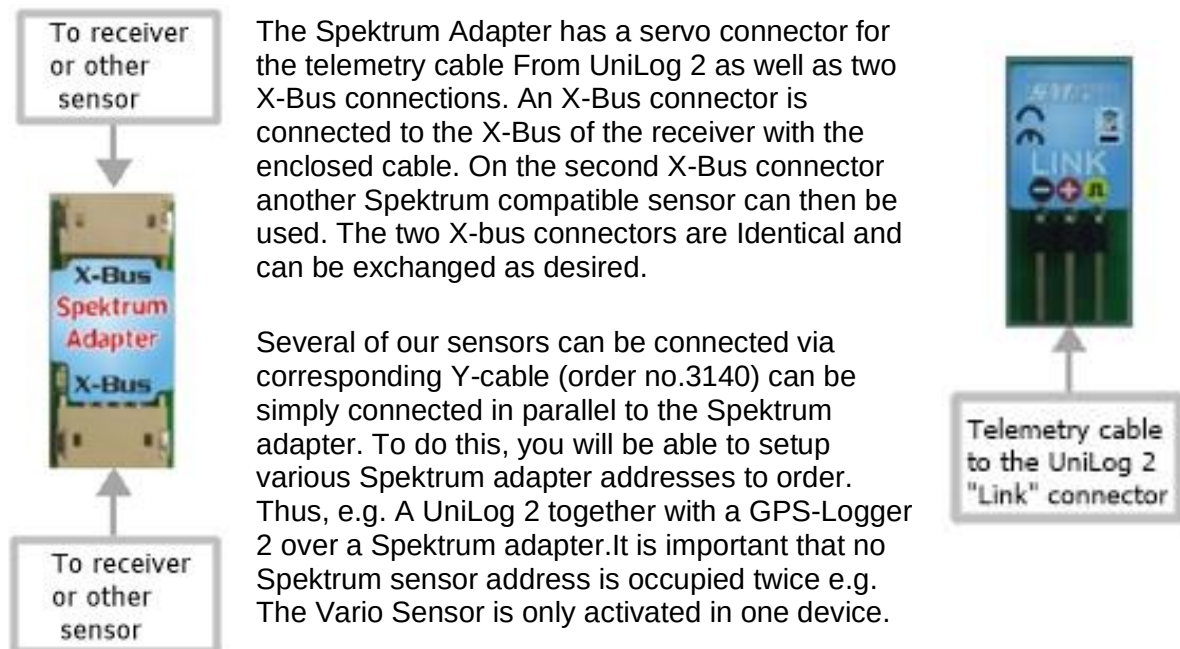
In principle, the alarms in the transmitter are defined by FrSky. The UniLog 2 has no way to directly alarm on the transmitter. All alarm thresholds as well as the varioton generation are thus specified in the transmitter.

9.7. Spektrum

As of firmware v1.13, UniLog 2 can also be used with Spektrum telemetry. The UniLog 2 equipped with our **Spektrum Adapter order no. 9120** is connected to the receiver. This adapter translates the telemetry values into the Spektrum X-Bus format.

9.7.1. Spektrum-Adapter

The Spektrum adapter is compatible with all telemetry-capable Spektrum receivers and with the TM1000 System. Currently the receivers with integrated telemetry send the data slightly slower than the TM1000, so there is some longer delay in the vario tones. Therefore, in the UniLog 2 set the Vario Filter to "fast".



Registration at the Spektrum Transmitter:

The query for the connected sensors with Spektrum comes only once after switching on the receiver. Here the Spektrum adapter responds immediately with the sensors it has learned during the last operation. This means that after every change in the configuration, i.e. after deleting or activating a

Spektrum sensor in the UniLog 2, the system must be first switched on to learn the new sensor types. The Spektrum adapter cannot be changed until the second power up sign in.

LED:

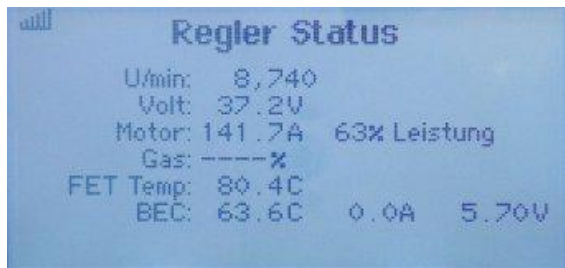
At start-up, the integrated LED goes slowly on and off. Then it flickers in operation, if data is sent by X-Bus.

Each time a sensor is registered or logged off, the LED goes slowly on and off to confirm the operation.

9.7.2. Presentation on the Transmitter

The UniLog 2 can represent a total of five different Spektrum sensor types. All can be individually deactivated to avoid double occupancy on the X-Bus.

The data can be further processed in the transmitter exactly as in the case of original Spektrum sensors, that is to say speech output, display, alarms and memory are possible.



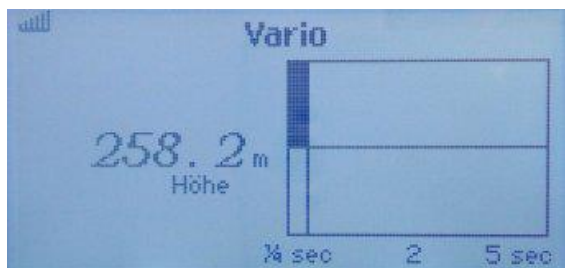
ESC Sensor ("Controller Status")

- rpm
- Volts
- Current
- Residual capacity in% as "% power"
- Temperature sensor at A1 as "FET Temp"
- Temperature sensor at A2 as "BEC Temp"
- Receiver voltage as "BEC V"



Power / mAh ("battery capacity")

- Current and max current
- Used capacity
- Temperature sensor on A3



Vario Sensor ("Vario")

- Height
- Vario



LiPo Monitor ("6s LiPo Monitor")

- Cells 1 to 6
- Temperature sensor on A3, if present, Or internal temperature



Airspeed

- This sensor is automatically used when a speed sensor is configured at an input A1 to A3

9.7.3. Alarms

In principle, the alarms are defined in the transmitter. The UniLog 2 has no way to directly trigger an alarm on the transmitter. All alarm thresholds and also the Vario tone generation are thus given in the transmitter.

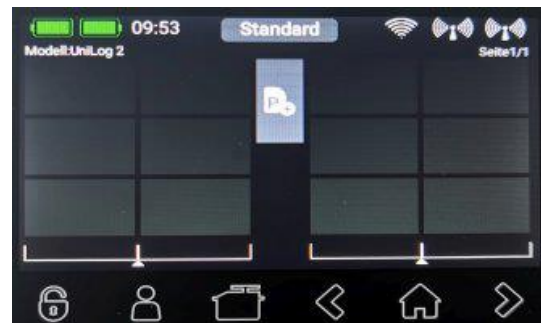
9.8. PowerBox Core P2Bus

From firmware v2.15 the UniLog 2 can also be used with the PowerBox Core Telemetry. by virtue of the limited storage space there is a separate firmware for the core, recognizable by the version number 2.xx. The UniLog 2 currently supports full transmission of all telemetry values and switching between German/English and meters/feet for the units. A setting menu of the UniLog 2 is not yet installed.

9.8.1. Presentation on the Transmitter



Empty start screen



A long press on the free area brings up the window selection.



After clicking on the desired window, with... "Telemetry" ...



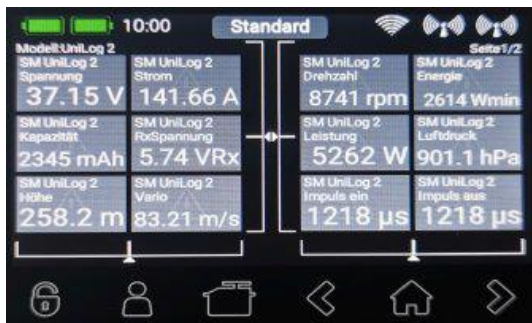
a widget will be created. The + opens the sensor list.



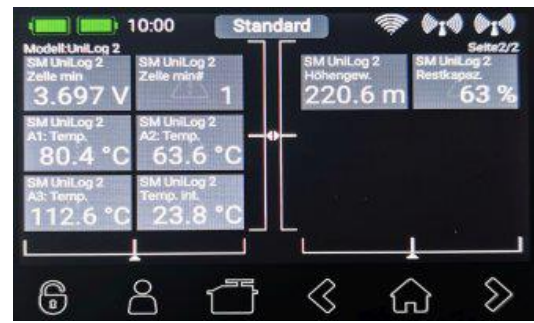
Now the desired value can be selected



This widget can also have multiple values displayed alternately, just insert with +.



A fully occupied telemetry site ..



... it continues on page 2.



Alarms are defined directly in this widget in the transmitter.



A long click on the sensor in the widget also Shows the address of the sensor. This can also with X → Y be changed to allow several similar sensors to be connected.



English designations and units after Changeover in the system. The sensors switched have to be rescanned in a sensor Widget.



The values can also be displayed in large format.

9.8.2. Alarms

In principle with PowerBox, the alarms are defined in the transmitter. The UniLog 2 has no way to directly trigger an alarm on the transmitter. All alarm thresholds and also the Vario tone generation are thus specified in the transmitter.

10. Data evaluation

Data is read from the memory card using the software LogView (www.logview.info) or with the GNU DataExplorer (www.nongnu.org/dataexplorer/index.en.html). You can use the software each time from this homepage free of charge.

10.1. GNU DataExplorer

The UniLog 2 is integrated as a separate device in the DataExplorer. Simply chose the “UniLog 2” in the device selection dialog, then you can use "import data" "directly import the desired file from the memory card

Of the UniLog 2. The software is currently being further developed and can also be used with many other devices and protocols from the RC area. The data can be displayed in the form of waveforms in DataExplorer and can be further extensively processed. Also, curve matching, attachment of notes etc. is possible. Please read the documentation for the program or visit the homepage.



Illustration 1: A short flight with an electrical model including speed sensor

10.2. LogView

The UniLog 2 is integrated into LogView as a separate device. Simply select the “UniLog 2” in device selection, then via “File” → “Import Device Data” you can import the desired file directly from the memory card of the UniLog 2.

The data can be displayed in graph form and also extensively processed in LogView. Also curve comparison, attaching notes, saving, etc. is possible. Please read the documentation for the program or get information on www.logview.info and/or in their user forum.

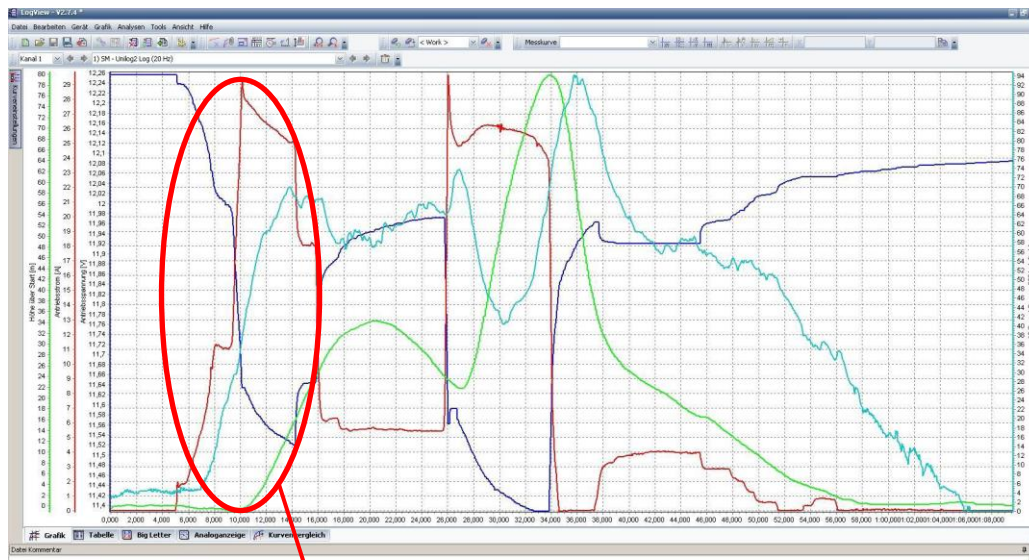
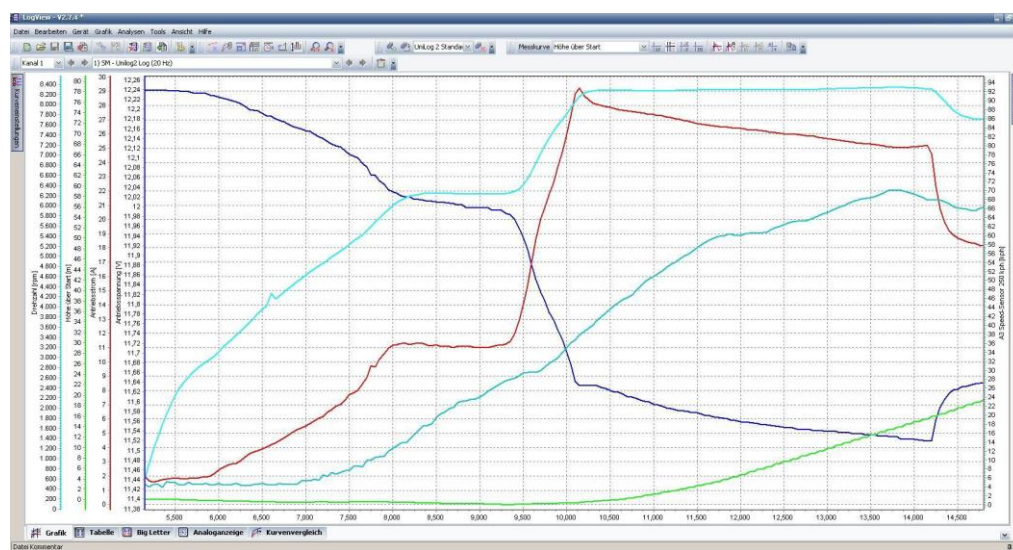
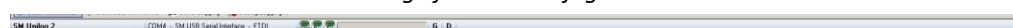


Illustration 2: A short flight with an electrical model including speed sensor



V Illustration 3: marked range from above flight zoomed in - at 20 Hz data rate



11. The software “SM UniLog 2 tools”

On our homepage you will find in [Software & Updates](#) the free software “SM UniLog 2 tools” with the following functions:

- Select and change UniLog 2 settings via the memory card
- Live data display over the USB interface (stock no. 2550) with a simulated UniDisplay
- automatic on-line search for new firmware for the UniLog 2 on our Internet site

If you move the mouse over the buttons, you will see short help texts for the operation.

11.1. Settings

The software opens with the “Settings” tab. Here the .ini file of the UniLog 2 memory card is read and the settings displayed. If a value is changed, the data must be written to the memory card with the “Sichern” (Save) button.

Illustration 3: marked range from above flight zoomed in - at 20 Hz data rate

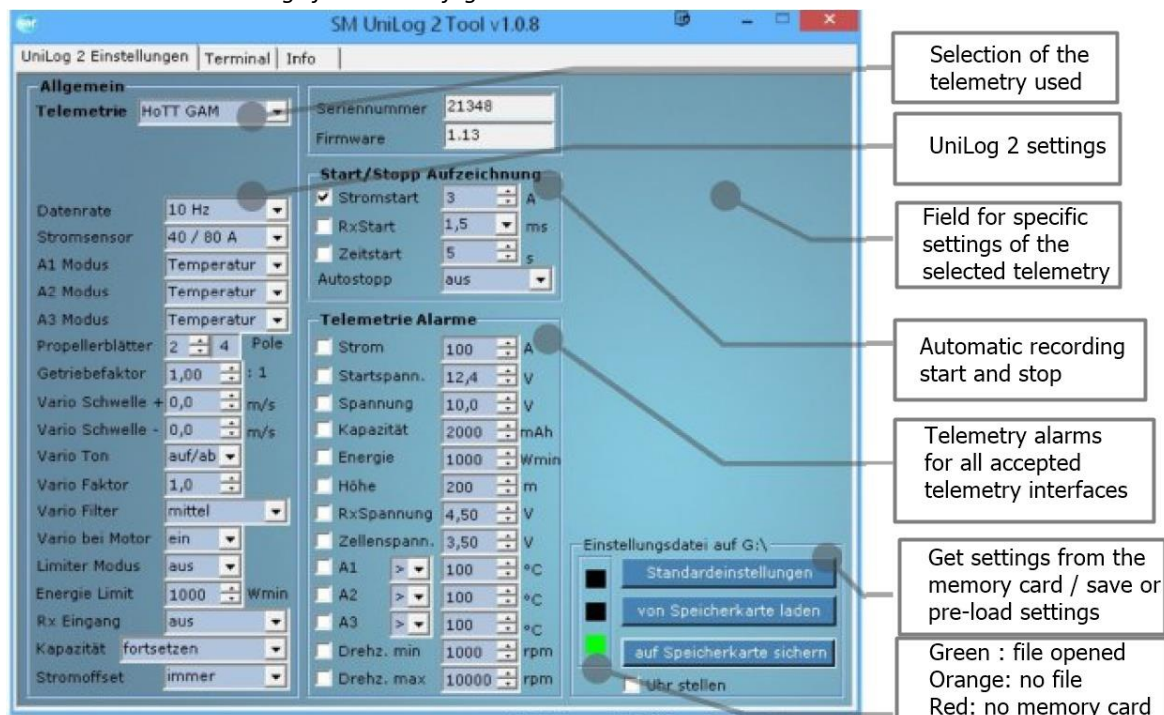


Illustration 4: Settings Tab

11.2. Live access to the UniLog 2

If the UniLog 2 is connected to a PC with the USB interface, it can be accessed via the Terminal Function of our software. The display is identical to the live operation of the UniDisplay. See also chapter 8.

But a download of the data from the memory card is not possible via the USB interface.

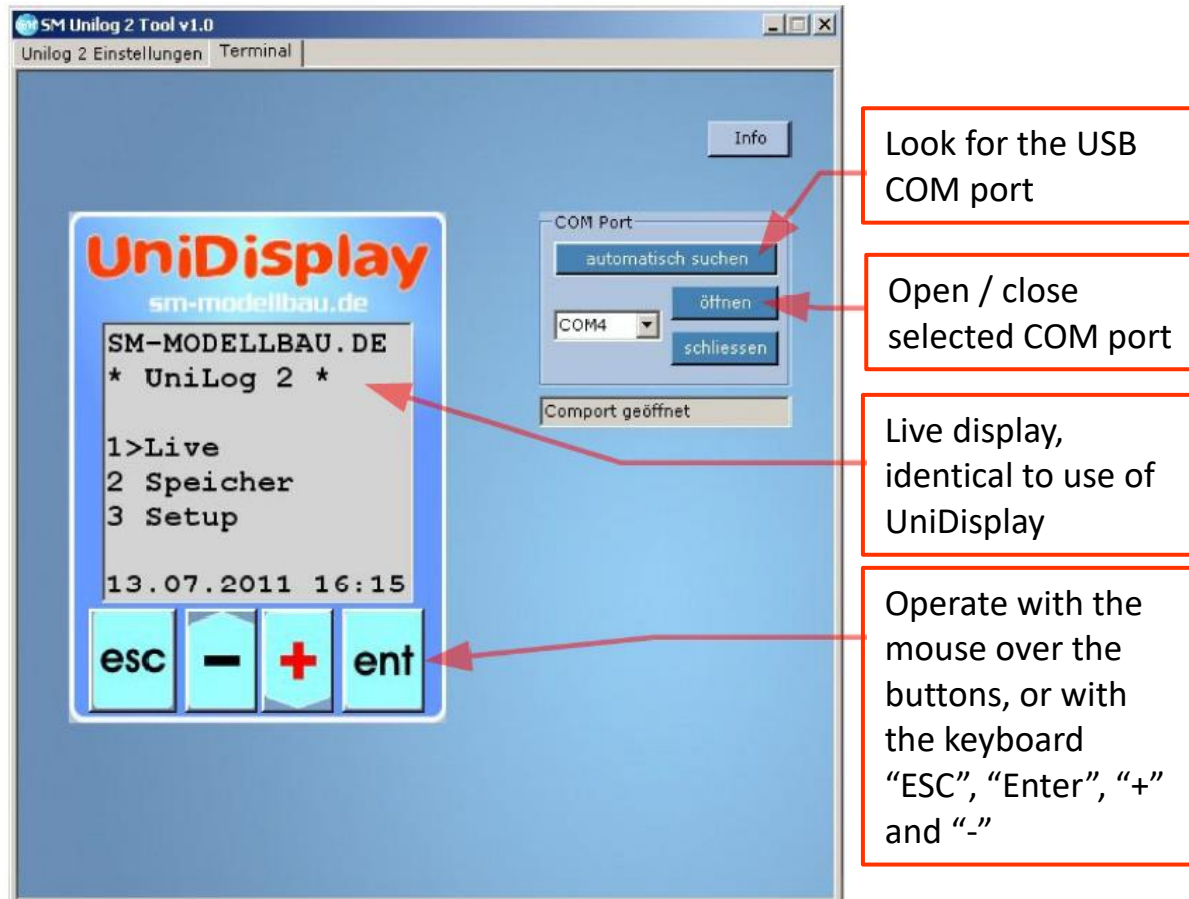


Illustration 5: Terminal tab

The USB interface is identical to the one used with the previous UniLog or with the GPS Logger 1/2. It can be ordered separately under the part no. 2550.

The UniLog 2 must be supplied with external power for connection to the PC!
For example, via a current sensor, or use a 4 cell Receiver battery at the "Link" connector.

12. Firmware update of the UniLog 2

New firmware is simply up-loaded to the UniLog 2 via the micro SD memory card. On improvements to the firmware an appropriate file with the update can be downloaded free of charge from our homepage www.sm-modellbau.de in the menu item **Software & Updates**.

When the PC software “SM UniLog 2 Tools” is started, the program automatically searches for a newer firmware on our server. If a newer file is found this can, on request, be automatically loaded on to the memory card. In this way the UniLog 2 always remains up to date with the newest version.

The file has, for example, the following file name for the version v1.00: **UL2FW100.UPD**

The file name of the firmware update must not be changed!

Procedure:

- The file is simply copied into the root directory of the memory card, directly into the main directory, not into a folder.
- At the next start of the UniLog 2 the red LED flashes during the update procedure.
- Then the UniLog 2 starts with the new firmware.
- The file must remain on the memory card, it is only loaded once.

13. Version history

Here you will find all firmware versions and the changes to the previous version.
You can find the current version of your UniLog 2 firmware with our software "SM UniLog 2 Tools" or the UniDisplay.

Version	Date	Comment
1.00	08-2011	First released version
1.01	10-2011	1 The battery capacity measurement can now be continued over several flights. If the same battery reconnected without charging the values from the last flight are used as starting capacity and energy. 2 In HoTT is now the value according to the settings for the Vario "Vario threshold" and "Vario Sound" modified, similar to the issue at M-Link. 3 In UniDisplay "Live" addition to the energy and the rpm per volt [kV] is displayed 4 Error on M-link recording fixed. 5 Error at startup "RX ein" fixed.
1.02	10-2011	1 The continued capacity measurement was at first use is not always correct initialized.
1.03	11-2011	1 The continued capacity measurement another error.
1.04	01-2012	1 Support for Graupner HoTT telemetry V4 (V3 is no longer supported in this release) 2 When the limiter is in operation, the capacity and energy is always reset at the start. 3 M-Link addresses are now checked on double occupancy. 4 Capacity alarm is cleared after 10 seconds, but then comes permanently once another 5% of the adjusted capacity were consumed.
1.05	06-2012	1 Integrated limiter for F5J 2 Vario now has separate thresholds for rising and falling 3 Jeti Duplex: EX installed telemetry 4 HoTT: Speed from speed sensor is transmitted only in the intended value, not more than "ml" 5 M-Link: cell voltages could not always be displayed, now as a "V" with a decimal point output, new value "cell min" for the lowest cell voltage 6 Improved accuracy of the internal clock 7 Installed alarms on the inputs A1 to A3
1.06	10-2012	1 Jeti Duplex : Varioton did not work on all systems 2 HoTT : - negative current , capacitance and voltage values led to meaningless indicators on the transmitter - TMP1 and TMP 2 show now up to 2 external temperature sensors , first on TMP1 , second on TMP2 , without or with only one sensor is TMP2 the internal temperature 3 Telemetry Alarms A1 to A3 could not be switched off 4 Fixed issue with many files on the memory card : it could possibly be a new file created and the newly launched UniLog 2 5 if only the balancer iand not a current sensor connected , the sum

		of the individual cells is used as battery voltage, the voltage alarms are active (this approach requires a separate ground connection to UniLog 2) 6 Recognition of the absence of ground connection for single cell measurement > all the cells show 0 7 Plugging in the drive battery can now also be done on the fly > consumption data is checked each time 8 capacity manual reset is now as when UniSens - E by switching off during the LED running light possible > Current offset is set only for the first time
1.07	11-2012	1 Speed sensor was set at the start with v1.06 no longer to 0 2 HoTT: UniLog 2 can now optionally as GAM (General module), EAM (Electric Air Module) or ESC (controller) that can be configured → together several UniLog 2 be operated on HoTT
1.08	01-2013	1 Jeti EX: Correction for Jeti Transmitter DC-16 firmware v1.07
1.09	04-2013	1 Futaba FASSTest added S.BUS2 telemetry 2 JR DMSS Telemetry added 3 continuous capacity measurement can be explicitly turned off and now 4 Recalibration of the current zero point when switching can now be switched off and be (current offset option = never / always) 5 new reading "height gain": change in height of the last 10 seconds, every second recalculated -> with Jeti EX and M-Link as a new value in HoTT as m/3s 6 while the battery detection speed measurement is suppressed here because otherwise incorrect values are measured by the beeping of the controller 7 Alert Height listen now on after 20 seconds and only when new crossing Triggered threshold again 8 HoTT: Log in ESC mode has been changed
1.10	05-2013	1 Jeti sensor name changed to "UniLg2": many texts fit better into the transmitter display 2 at "current offset never" the battery has initialization does not work, so no speed measurement and no battery alarms.
1.11	09-2013	1 FASSTest: protocol of the current sensor 1678 installed → the UniLog 2 sends the data now in this format with V, A and mAh → the current display only goes up to 160 A, so the current / 10 is transmitted at 400 A Sensor: 400 A are on the display 40.0 A NOTE: The sensor must be removed from the transmitter and then re-registered! 2 Speed sensor 600/1000 km / h built as a special version 3 Speed sensor offset is set also subsequently 5 seconds after you connect the speed sensor 4 Telemetry detection did not work when the drive was activated before the receiver battery supply 5 Undervoltage alarm now triggers only after 2 seconds.

1.12	05-2016	1. Jeti: - Adaptation to REX receivers - transferred values can be selected individually 2. HoTT: - Text mode was not available for some recipients - UniLog 2 can send data as "HoTT Vario" 3. Futaba: Inputs A1 to A3 register as TRUE AIRSPEED SENSOR NO. F1677, if a speed sensor is configured there. 4. FrSky telemetry installed 5. No more automatic detection of telemetry at HoTT / Jeti / MPX, now needs to be set manually. 6. new option "fixed serial number" sends serial number 12456 to Jeti and Futaba 7. Display of the energy in Wmin for all telemetries: As "ml", Futaba as SBS01T, Jeti as "Energy", FrSky as "Fuel ml", M-Link as "ml", JR DMSS as power "W" (only with limiter F5B / F / D) 8. New telemetry Energy alarm 9. Calculation of the Vario optimized and accelerated: - Vario values are now less smoothed but are more up-to-date - Transfer of the Vario values faster especially with Jeti and HoTT - new option "VarioFilter": "slow" corresponds approximately to the previous evaluation, "medium" And "fast" are correspondingly faster - Height display is faster 10. When controlling of the Min/Live/Max display via Rx The minimum and maximum values can be now be reset by speedy three times operation. 11. Error in height measurement at recording start corrected 12. Rx input can now also be used as a rpm input with galvanic isolation (for ignition system signals) The rpm input is then automatically Rx input (inputs are exchanged) 13. Min alarm on A1 to A3 (e.g., speed sensor) is output only for 10 seconds; then only after the Min value has been exceeded 14. Option "Set Clock" via UniLog 2 Tool: if activated the time of the ini file at the Next start set as internal time.
1.13	02-2017	1. Spektrum telemetry installed via Spektrum adapter: - up to 5 sensors: ESC sensor, current / mAh, VarioSensor, LiPo monitor, Airspeed 2. M-Link: not all addresses could be set via the UniDisplay 3. Futaba: - Compatibility mode for Robbe telemetry box - the signal quality output from the receiver is read in and set to (25/50/75/100) over an additional sensor slot as TEMP125 4. FrSky: Cells Display has no Sum, Diff, Min and Max displayed 5. Telemetry alarm current will only turn off after 1 second 6. Alarms installed for minimum and maximum speed 7. Gently zero the height when the model is on the ground 8. Residual capacity as % value from alarm capacity and consumption on all telemetries except JR 9. Manual reset of the maximum values via the switching channel is always set for the option "Capacity = Continue" to the capacity 0 if the model was not yet in the air.

1.14	05-2017	1. Spectrum: negative values, e.g. ESC current, cause a display error and are now suppressed 2. HoTT: Alarm 'E' on the mc-28 no longer produces "pling-pling" but "weak telemetry channel" → previous alarm 'E' for alarms without corresponding voice output deleted without replacement → COM plug in and unplug, start and stop saving, A1,2,3 in millivolt mode 3. Futaba: when displaying the signal quality, the values 50% and 75% were reversed.
1.15	02-2020	1. Core (P2BUS) support built-in: - separate firmware with version number 2.xx - currently only display of measured values, assignment of address and switchover German/English and Metric/Feet, no menu - Default address: 187 2. gentle zeroing of the height can now be switched off 3. F3G limiter built-in 4. F5J Limiter: Motor can be switched on again, but leads to zero evaluation "--.-m 5 Jeti: Telemetry data is now sent faster at startup so that the receiver cannot briefly go into the range test 6. Jeti: height and height gain are displayed from 10 m without decimal point 7. HoTT: A1-A3 alarms for min. speed are output as "minimum speed".



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